Dohyun (Eric) Kim

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PROFILE

I'm a Penultimate Computer Systems Engineering (Hons) student at the University of Auckland, passionate about embedded systems and the hardware-software boundary. I bring full-stack hardware experience from FPGA to PCB, a competition-winning project portfolio, and AWS certifications.

EDUCATION

University of Auckland - Auckland CBD, New Zealand *Expected Graduation Nov 2027*

- Degree: *Bachelor's (Hons) of Engineering in Computer Systems Engineering*
- Concentration: Embedded Systems & Digital Hardware Design

EXPERIENCE

Teaching Assistant **Auckland, New Zealand**
University of Auckland *Jun 2026 - Present*

- Delivered teaching support across tutorials, lab sessions, and test invigilation for **ELECTENG 292: Electronics** to strengthen **student leadership** and technical laboratory techniques.

Robotics Instructor **Auckland, New Zealand**
Creative Imaginary Lab (ciLab) *Apr 2026 - Present*

- Instructed students in robot **hardware assembly and software programming** to complete missions, while coaching and supporting teams in preparation for **nationwide robotics competitions**.

Front of House (FoH) **Auckland, New Zealand**
Twelve Restaurant *Jul 2024 - Jan 2025*

- Collaborated with the team to keep front-of-house running smoothly, resolving complex customer situations efficiently.

PROJECTS

Flappy Universe - VHDL | Team Project | 2026

- Implemented a **Flappy Bird-style game** in VHDL on an **Altera FPGA** with VGA signal generation, PS/2 mouse input, sprite rendering from ROM, and an LFSR random number generator.
- Developed a **layered graphics pipeline** using modular VHDL components and a pixel-priority compositing system to render animated game scenes at VGA resolution in real time.

Smart Energy Monitor - C & Embedded | Team Project | 2025

- Designed an embedded **energy monitoring system** using ATmega microcontrollers with sensor interfacing, ADC processing, and signal conditioning **to measure and display real-time energy usage**.
- Produced a validated PCB schematic using **Altium Designer and LTspice simulations** to achieve accurate sensor-to-display signal conditioning across the full hardware stack.

AWARDS

3rd Place in 2025 ECSE Design Competition **Team Project**
"Winnie the Bot" - Interactive companion robot *Sep 2025*

- Built an AI-powered robot using dual **ATmega328P** microcontrollers, servos, an AI camera, and audio peripherals, in order to enable face tracking, arm movement, and voice dialogue.
- Contributed the **embedded hardware** and **enclosure prototyping**, in order to fit all electronics into a compact form.

CERTIFICATIONS

Amazon Web Services (AWS) Certified **Cloud Practitioner** *Apr 2026*

- Demonstrated foundational knowledge of **AWS cloud concepts**, core services, and cloud architecture best practices.

Amazon Web Services (AWS) Certified **AI Practitioner** *May 2026*

- Demonstrated knowledge of AI and machine learning concepts, **AWS AI/ML services**, and responsible AI practices.

ACTIVITIES & LEADERSHIP

Academic Team Executive **Auckland, New Zealand**
KEB (Korean Engineering Body) *Jul 2024 - Present*

- Tutored junior engineering students and assisted in planning academic events for the student community.

Full time Volunteer **Auckland, New Zealand**
IEEE (Institute of Electrical and Electronics Engineers) *Jul 2025*

- Volunteered 40+ hours full-time supporting competition operations at the nationwide NZ Robotics Olympiad 2025.

SKILLS

Hardware & Embedded: C, VHDL, MATLAB, Atmel AVR, Altium Designer, LTspice, Proteus, Intel Quartus, AutoCAD
Software & Cloud: Python, Java, JavaScript, React.js, Pandas, Scikit-Learn, AWS, Git/GitHub